

Homework 4

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STAT 7630 — Homework: Variable Selection, Ridge Regression, and Random Effects

Instructions: Show all work for derivations. Provide clear interpretations in words where requested. You may use R for computational parts.

Problem 1 — Variable Selection and Model Comparison (Conceptual + Applied)

You are analyzing a dataset with response Y and predictors X_1, X_2, X_3, X_4 . You consider the following models:

- $M_1 : Y \sim X_1 + X_2$
- $M_2 : Y \sim X_1 + X_2 + X_3$
- $M_3 : Y \sim X_1 + X_2 + X_3 + X_4$

The following AIC values are obtained:

Model	AIC
M_1	210
M_2	205
M_3	204

(a)

Compute ΔAIC for each model.

(b)

Which model would you select? Is there strong evidence against any model?

(c)

Explain why selecting M_3 may not necessarily be the best decision for prediction.

(d)

Explain the difference between using AIC and a partial F-test for model selection.

Problem 2 — Ridge Regression Derivation

Consider the ridge regression estimator:

$$\hat{\beta}_{ridge} = \arg \min_{\beta} \{ \|y - X\beta\|^2 + \lambda \|\beta\|^2 \}$$

(a)

Derive the closed-form solution for $\hat{\beta}_{ridge}$.

(b)

Explain why $X^T X + \lambda I$ is always invertible when $\lambda > 0$.

(c)

Explain how ridge regression modifies the OLS estimator and why this improves stability.

Problem 3 — Ridge Regression and Bias–Variance Tradeoff

(a)

Explain why ridge regression is biased. What direction is the bias?

(b)

Explain why ridge regression typically has lower variance than OLS.

(c)

Under what conditions will ridge regression outperform OLS in terms of mean squared error?

(d)

Give an example of a situation where ridge regression may perform worse than OLS.

Problem 4 — LASSO and Variable Selection (Conceptual + R)

(a)

Write the LASSO optimization problem and compare it to ridge regression.

(b)

Explain geometrically why LASSO can set coefficients exactly equal to zero.

(c) (R)

Simulate a dataset with: - $n = 100$, $p = 10$ - only 3 predictors having nonzero coefficients

Fit: - OLS

- ridge regression

- LASSO

Report: - estimated coefficients

- which variables LASSO selects

(d)

Explain why LASSO may select different variables across repeated samples.

Problem 5 — Random Effects Model: Mean and Variance Structure

Consider the random intercept model:

$$Y_{ij} = \beta_0 + \beta_1 X_{ij} + u_i + \varepsilon_{ij}, \quad u_i \sim N(0, \sigma_u^2), \quad \varepsilon_{ij} \sim N(0, \sigma^2)$$

(a)

Compute $E[Y_{ij} \mid X_{ij}]$.

(b)

Compute $\text{Var}(Y_{ij})$.

(c)

Compute $\text{Cov}(Y_{ij}, Y_{ik})$ for $j \neq k$.

(d)

Derive the intraclass correlation coefficient (ICC) and interpret it.

Problem 6 — Random Effects vs Fixed Effects (Conceptual + Applied)

Suppose you are analyzing test scores of students within schools.

(a)

Write a fixed effects model for this problem.

(b)

Write a random effects model.

(c)

Explain the key conceptual difference between the two models.

(d)

Under what conditions would you prefer a random effects model?

(e) (R)

Simulate a dataset with: - 20 groups (schools)
- 10 observations per group

Fit: - a standard linear model ignoring grouping
- a random intercept model using `lme4::lmer()`

Compare: - coefficient estimates
- interpretation of group effects

Problem 7

Weisberg Problem 10.4